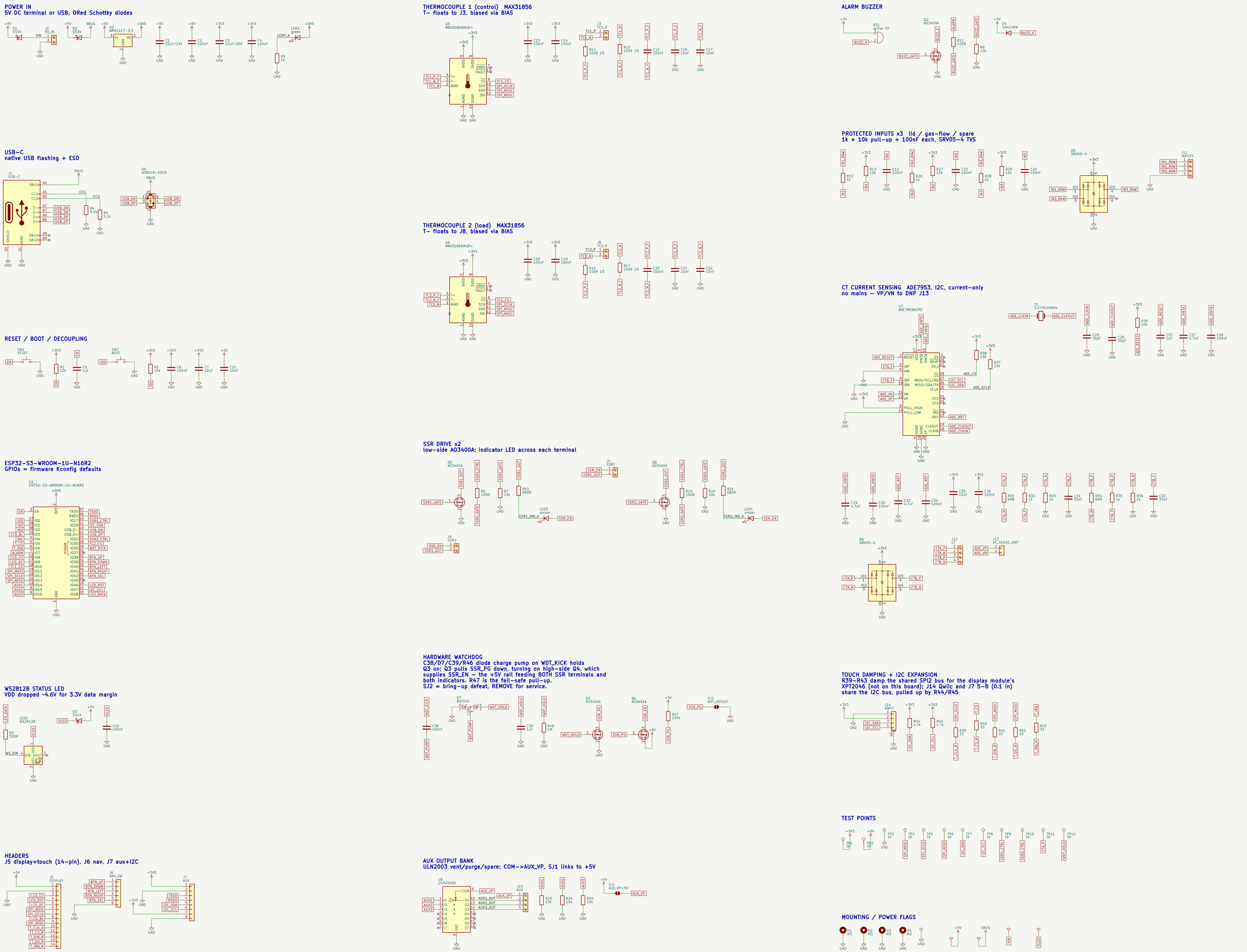

NOTES – Bisque kiln controller, rev B

SSR terminals J4/J9: pin1 = SSR_EN (board +5V, watchdog-gated), pin2 = the switched low side. Boot-safe: R7/R20 hold each MOSFET gate down while the ESP32 pins are high-impedance. NOT opto-isolated – rev B tried that and reverted it: an opto only isolates if the SSR loop is powered off-board, and this board powers it.

TC1 terminal J3 / TC2 terminal J8: pin1 = K+, pin2 = K- – both float, biased near AGND only through each MAX31856's internal BIAS network. Ungrounded-junction probes required: two grounded-junction probes in one kiln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND – dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1=+5V 2=GND 3=CS 4=RST 5=DC 6=SDI/MOSI 7=SCK 8=BL 9=SDO/MISO 10=T_CLK 11=T_CS 12=T_DIN 13=T_DO 14=T_IRQ. PIN 1 IS +5V, NOT 3V3 – the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31856s (R39–R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N – one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted – do not wire to AC mains. Y1/C25/C26 load caps are an ASSUMED value (unverified C_L), see design.py comment. ADE7953 I2C address 0x38 collides with PCF8574A.

4-layer, 100 x 100 mm, JLCPCB standard process
ESP32-S3-WROOM-1U-N16R2 + 2x MAX31856 + dual SSR + ADE7953
Bisque project

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Title: Bisque Kiln Controller

Size: A1	Date: 2026-07-20	Rev: B
KiCad E.D.A. 10.0.5		Id: 1/1